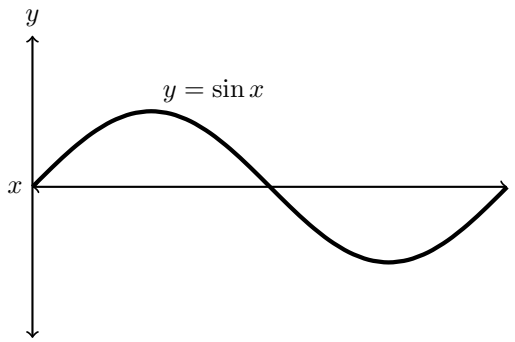
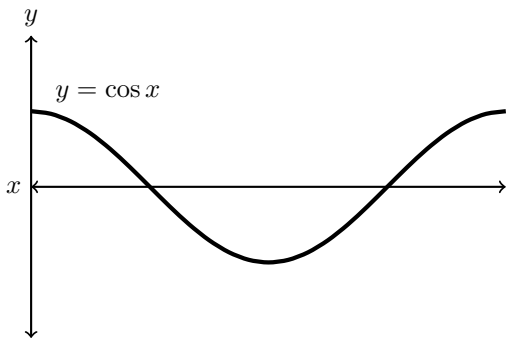
MATH 195: 11/4
GRAPHS OF TRIG AND INVERSE TRIG

Graphs of $\sin x$ and $\cos x$.

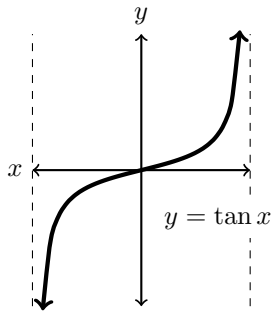
Both $\sin x$ and $\cos x$ have a period of 2π, a maxium of 1, and a minimum of -1.

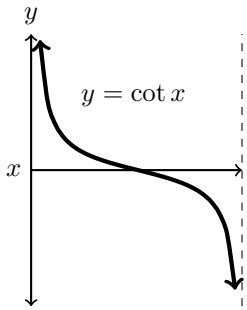




Graphs of $\tan x$ and $\cot x$.

$\tan x$ and $\cot x$ each have a period of π with asymptotes at the ends of each period. The asymptotes are placed in different locations: at the top/bottom of the unit circle for $\tan x$ and at the left/right of the unit circle for $\cot x$.

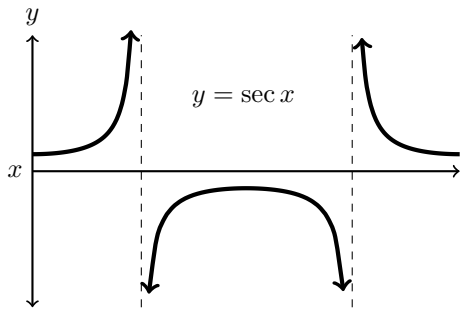


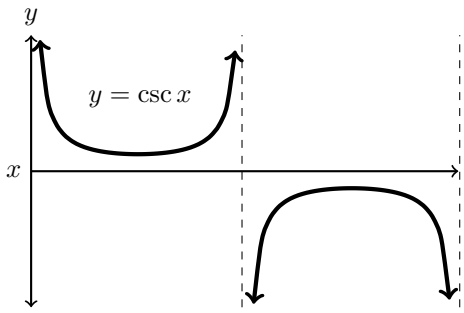


Writing the domain of tan, cot, sec, and csc in interval notion is annoying. Each of these has infinitely many vertical asymptotes, which means infinitely many holes in the domain, which means the domain is a union of infinitely many intervals. The way I advise you to think about it is in terms of the unit circle. Are the asymptotes at the top/bottom or left/right of the circle? If you know this, you can figure out the coordinates of asymptotes.

Graphs of $\sec x$ and $\csc x$.

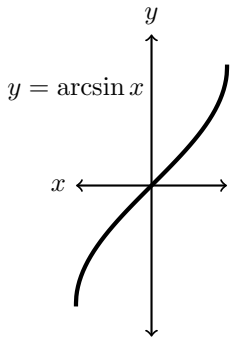
These look like "inside out" versions of the graphs of cos and sin, with asymptotes where their reciprocals of zeroes.

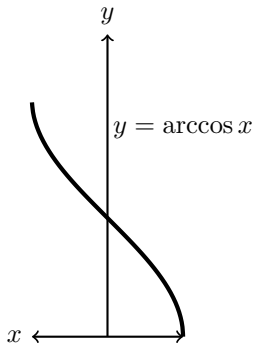


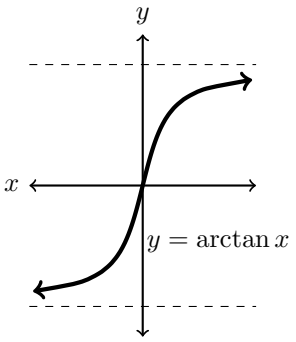


Graphs of inverse trig functions.

The arc-trig functions are defined by restricting the trig functions to a domain on which they are one-to-one, and then taking the inverse. Their graphs are thus diagonal reflections of a piece of a trig function—half a period for cos and sin, and a full period for tan.







arccos outputs an angle ot the top half of the circle $[0, \pi]$, arcsin outputs an angle on the right half of the circle $[-\pi/2, \pi/2]$, and arctan outputs an angle on the right half of the circle $(-pi/2, \pi/2)$, not including the top/bottom.

These pictures are of limited utility. Instead, I would advise you think of these in terms of the unit circle picture. arccos tells you which angle (on the top half) has a given x-coordinate, arcsin tells you which angle (on the right half) has a given y-coordinate, and arctan tells you which angle (on the right half) has a given slope.

(1) State the domain and range of arcsin.
(2) State the domain and range of arccos.
(3) State the domain and range of arctan.

The trig functions are periodic, so the outputs of the arc-trig functions are not the only angles with that x-coordinate/y-coordinate/slope. You can find the others by thinking geometrically with the unit circle picture. For these, let α be the angle output by the arc-trig function.

- For cos, do a vertical reflection to get the bottom half angle with that x-coordinate. In symbols, the angle is $-\alpha$. Then add integer multiples of 2π to get the general solution.
- For sin, do a horizontal reflection to get the left half angle with that y-coordinate. In symbols, the angle is $\pi - \alpha$. Then add integer multiples of 2π to get the general solution.
- For tan, rotate by a half circle to get the left half angle with that slope. In symbols, the angle is $\pi + \alpha$. Then add integers multiples of 2π to get the general solution.
  There is a shortcut: add integer multiples of π to α to get the general solution.